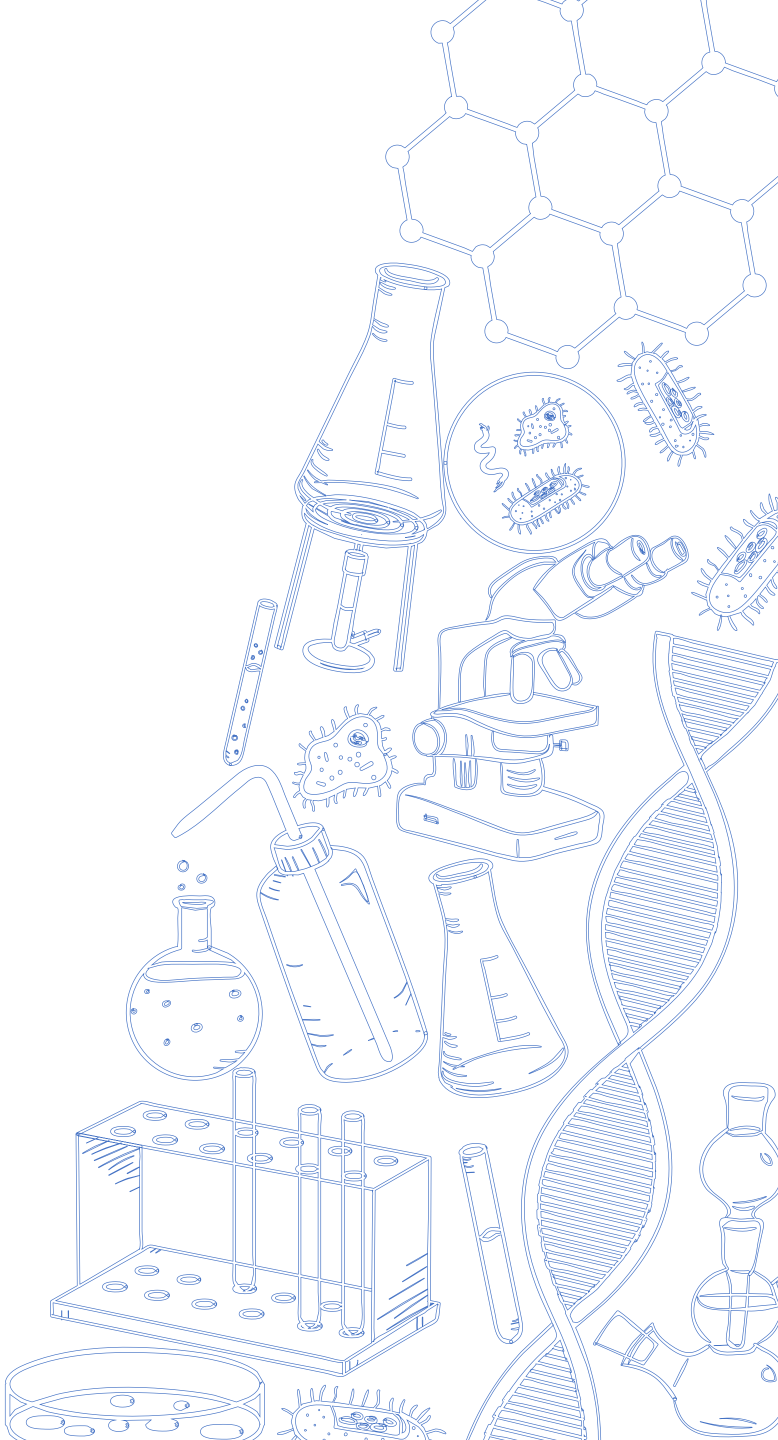
Science G10

Chemical Reactions

Class 2

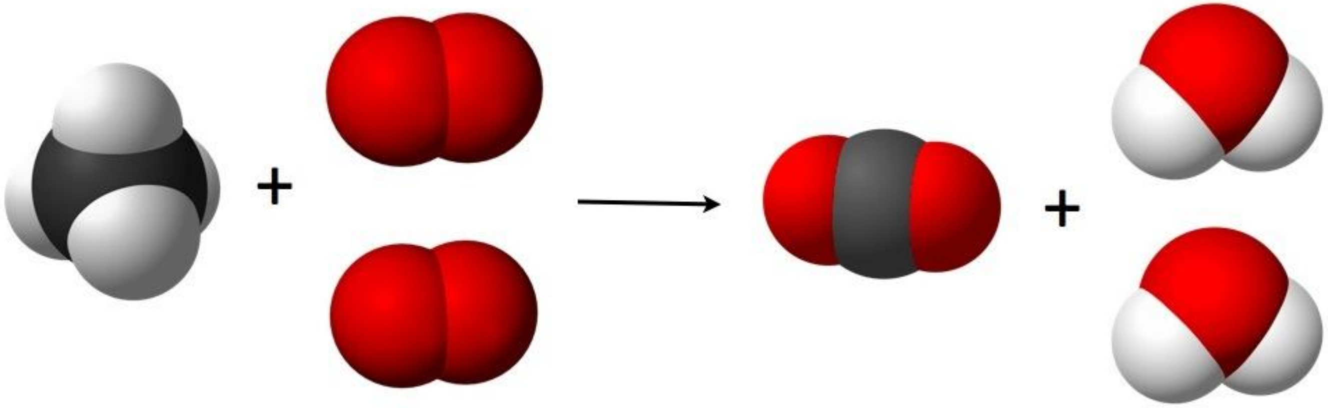
Agenda

1. Quiz 2
2. Lesson 2
   - Law of Conservation of Mass
   - Balancing Chemical Reactions
   - Types of Chemical Reactions
     - Synthesis
     - Decomposition
     - Combustion
     - Single Displacement
     - Double Displacement



Law of Conservation of Mass

- In any given reaction, the total mass of the reactants equals the total mass of the products
- Atoms cannot be created or destroyed



$$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$$

1 4   4   1 2   4 2 = 4

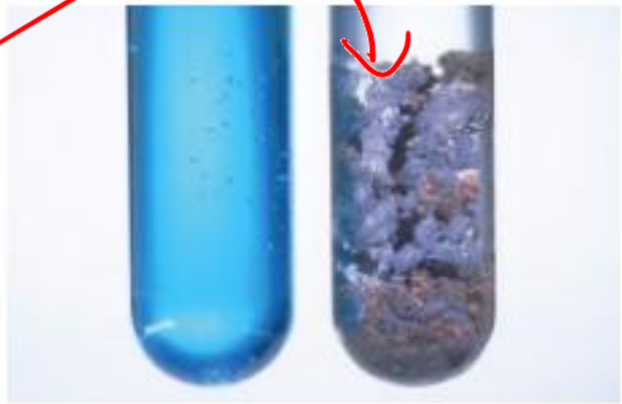
Chemical Reactions

REACTANTS	PRODUCTS

results

$Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + \boxed{Cu(s)}$

1 Zn	1 Zn
1 Cu	1 Cu
1 (SO_4)	1 (SO_4)



single displacement reaction

HCl hydrogen chloride

$HCl_{(aq)}$ Hydrochloric acid

State Symbols

important when dealing with acids

- State symbols are often written behind the chemical formula to indicate the state of the substance

$$Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$$

NaCl(aq)

State Symbol	Meaning
(s)	Solid
(l)	Liquid/Molten
(g)	Gaseous
(aq)	Aqueous (dissolved in water)

invisible

Fe_2O_3

Balancing Chemical Reactions

Fe^{3+} O^{2-}

Word Equation	Iron reacts with oxygen to form iron(III) oxide
Skeleton Equation	$4Fe + 3O_2 \rightarrow 2Fe_2O_3$
skip **Number of Atoms**	1 Fe 2 Fe 2 O 3 O
Balanced Reaction (add coefficients)	$\mathbf{4}\ Fe + \mathbf{3}\ O_2 \rightarrow \mathbf{2}\ Fe_2O_3$ You CANNOT change the subscripts when you are balancing. Only you can add or change the coefficients.

diatomic

multivalent ionic compound

2 1.5 1

x 2

4 3 2

$1.5\ H_2O \longrightarrow 3\ H_2O$

General Rules for Balancing Chemical Reactions:

- Only add coefficients, do not add subscripts
- ~~Balance non-hydrogens and non-oxygens first~~
- Balance polyatomic ions as a unit

H_3O_2

- All coefficients in the balanced chemical reaction need to be in simplest form as positive integers

no $H_{1.5}O$

- If chemical reaction cannot be balanced, check if the crisscrossing was done correctly

- always start with the atom with the least terms, if =1, you pick

STOP Checkpoint STOP

Balance the following chemical reactions:

a) 2 $K_2O \rightarrow$ 4 K + __ O_2

b) 2 $KClO_3 \rightarrow$ 2 KCl + 3 O_2

c) 3 $AlCl_3$ + 3 $Na_2CO_3 \rightarrow$ 1 $Al_2(CO_3)_3$ + 6 $NaCl$

d) 2 $NaOH$ + __ $H_2SO_4 \rightarrow$ __ Na_2SO_4 + 2 H_2O

2 H(OH)

e) 2 $N_2O_5 \rightarrow$ 2 N_2O_4 + __ O_2

Li + ~~H2O~~

HOH

? LiOH

H_2

Checkpoint

$Ag^{+}NO_3^{-}$

$AgNO_3$

Write the balanced chemical reaction of:

$Zn(NO_3)_2$

a) Magnesium with oxygen

$$2Mg + O_2 \longrightarrow Mg^{2+}O^{2-} = 0 \quad 2MgO$$

b) Methane (CH_4) burns in oxygen to produce carbon dioxide and water

$$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$$

ide

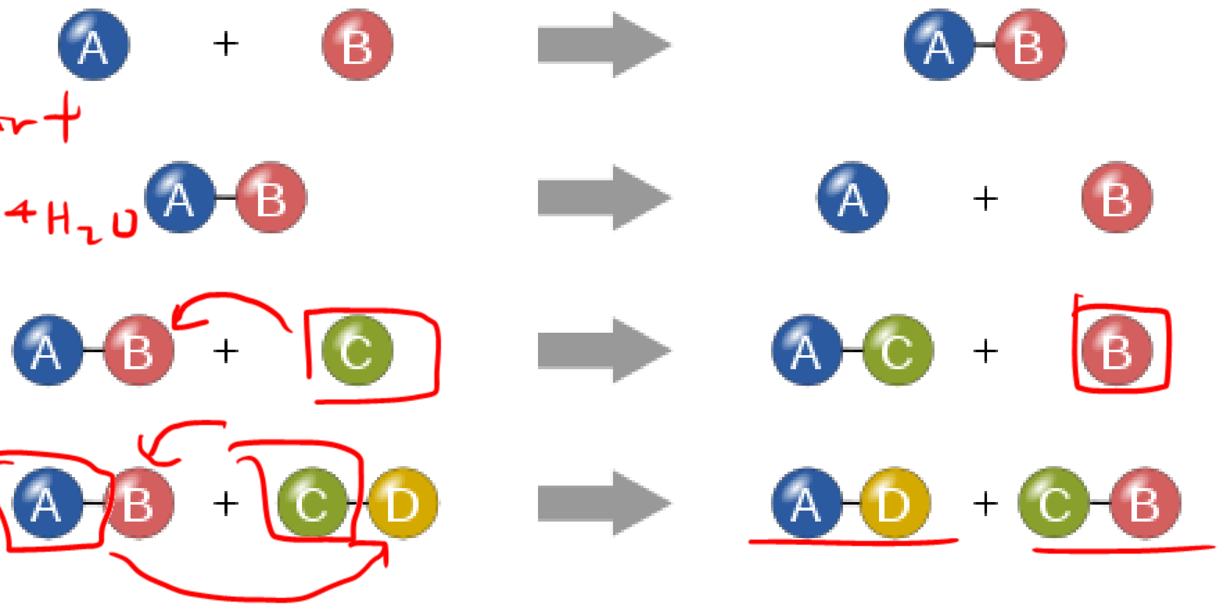
c) Zinc metal reacts in silver nitrate solution to produce zinc nitrate and silver metal

$$Zn + 2AgNO_3 \longrightarrow Zn(NO_3)_2 + 2Ag$$

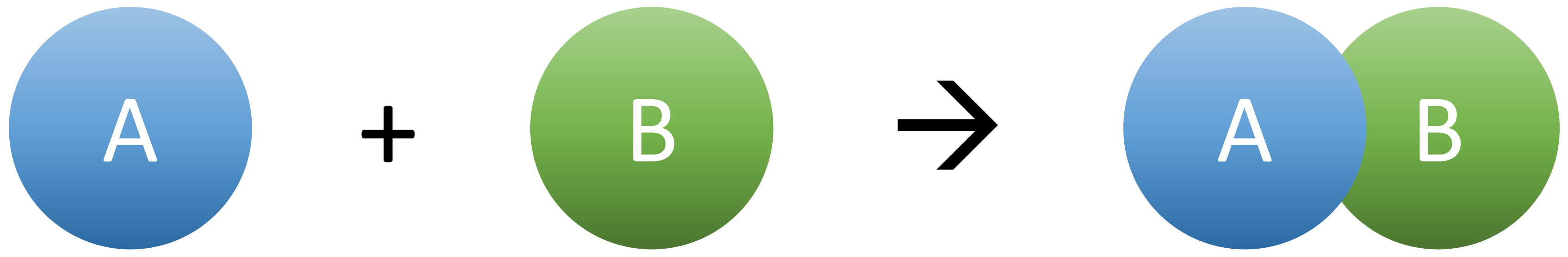
5g 1 ?g 2

Types of Chemical Reactions

- Synthesis combine
- Decomposition break apart
- * Combustion $x + O_2 \rightarrow CO_2 + H_2O$
- Single Displacement
- Double Displacement
  - Neutralization



Synthesis



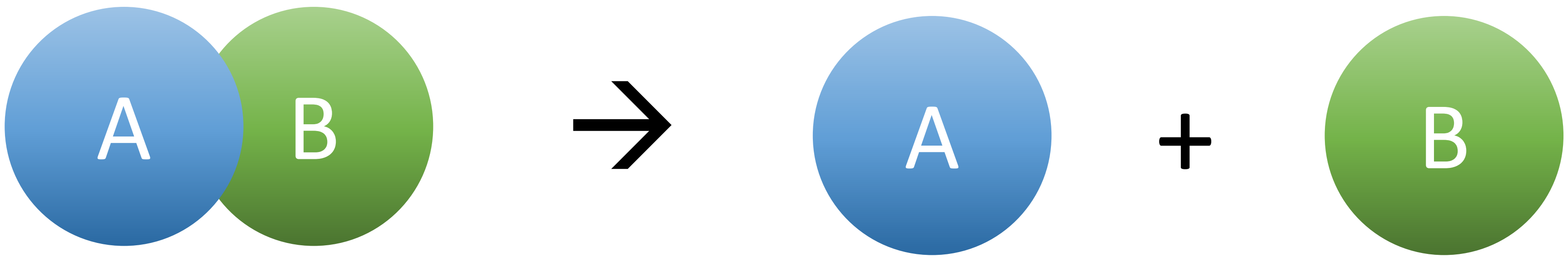
Examples:

$$3\overset{6}{H}_2 + \overset{2}{N}_2 \rightarrow 2\overset{2}{N}\overset{6}{H}_3$$

$$\overset{1}{Mg}\overset{1}{O} + \overset{1}{C}\overset{2}{O}_2 \rightarrow \overset{1}{Mg}\overset{1}{C}\overset{3}{O}_3$$

$$2CO + O_2 \rightarrow 2CO_2$$

Decomposition



Examples:

$$2\overset{4}{H}_2\overset{2}{O} \rightarrow 2\overset{4}{H}_2 + \overset{2}{O}_2$$

$$2Fe(OH)_3 \rightarrow Fe_2O_3 + 3H_2O$$

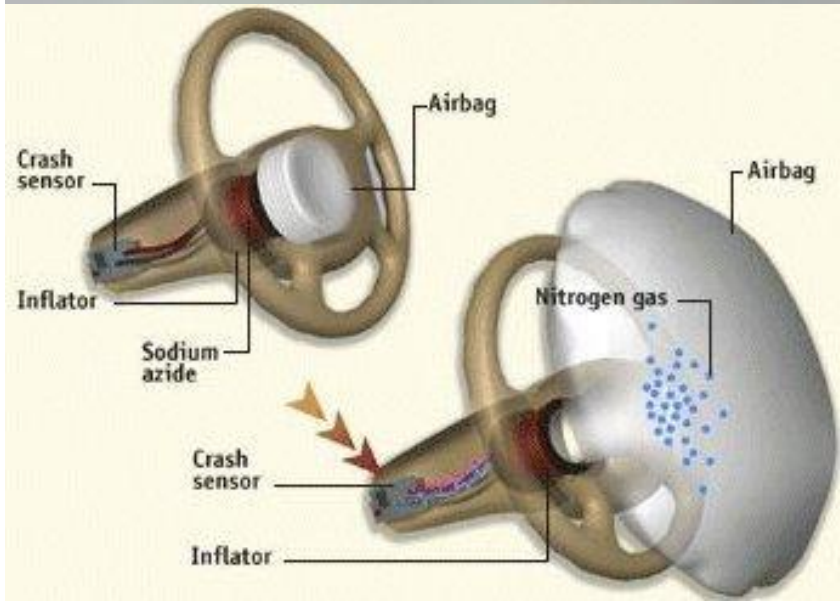
$$H_2CO_3 \rightarrow CO_2 + H_2O$$

Decomposition of Sodium Azide

$$2NaN_3(s) \rightarrow \underline{2Na}(s) + 3N_2(g)$$

? danger

- Airbags contain sodium azide, $NaN_3(s)$
- During a collision, electrical energy triggers the rapid decomposition of $NaN_3(s)$ into nitrogen gas to inflate the airbag



Checkpoint

Write the balanced chemical equation for the following and classify the type of reaction:

a) Zinc chloride → zinc + chlorine

$$ZnCl_2 \longrightarrow Zn + Cl_2$$

b) Potassium oxide + water → potassium hydroxide

$$\underset{2}{K}_2\underset{1}{O} + \underset{2}{H_2}\underset{1}{O} \longrightarrow 2\underset{2}{K}(\underset{2}{O}\underset{2}{H})$$

Combustion

- **Combustion** is a reaction in which a substance reacts with oxygen gas, releasing energy in the form of light and heat
- Two types:
  - **Complete combustion** occurs when there is sufficient oxygen
  - **Incomplete combustion** occurs where there is insufficient oxygen

C_2H_6 CH_3OH

CH_4

Complete Combustion of Hydrocarbons

- Hydrocarbons are compounds that contain carbons and hydrogens
- Complete combustion of hydrocarbons produces CO_2, H_2O, and energy
- Flame is blue in colour with little smoke

Complete combustion
Methane

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O + \text{energy}$$

Air Hole Open

STOP Checkpoint STOP

Assuming complete combustion, write the balanced chemical reaction for the combustion of:

a) Propane (C_3H_8)

$1\ C_3H_8 + 5\ O_2 \longrightarrow 3\ CO_2 + 4\ H_2O$

3 8 10 3 6 8 4

b) Octane (C_8H_{18})

$2\ C_8H_{18} + 25\ O_2 \longrightarrow 16\ CO_2 + 18\ H_2O$

16 36 50 16 32 36 18

32 + 18 = 50

Incomplete Combustion of Hydrocarbons

- Incomplete combustion of hydrocarbons produces CO_2, CO, H_2O, $C(s)$, and energy
- Flames from incomplete combustion tend to be yellow or orange in colour and are smokier due to unburned carbon

Air Hole Closed

$$C_4H_{10} + 5O_2 \rightarrow 2CO_2 + 5H_2O + \boxed{CO +} C(s) + \text{energy}$$

carbon monoxide

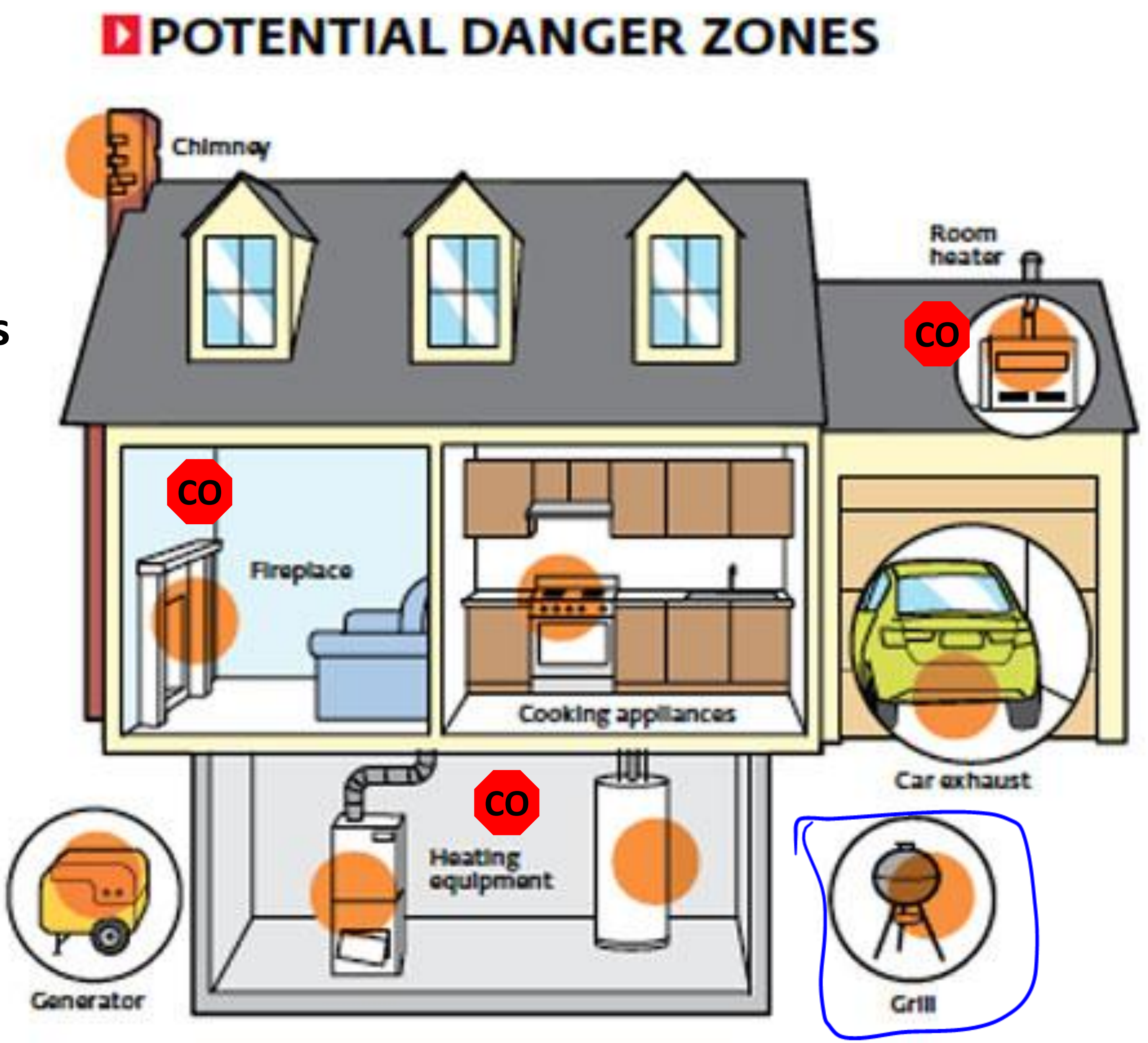
Dangers of Carbon Monoxide

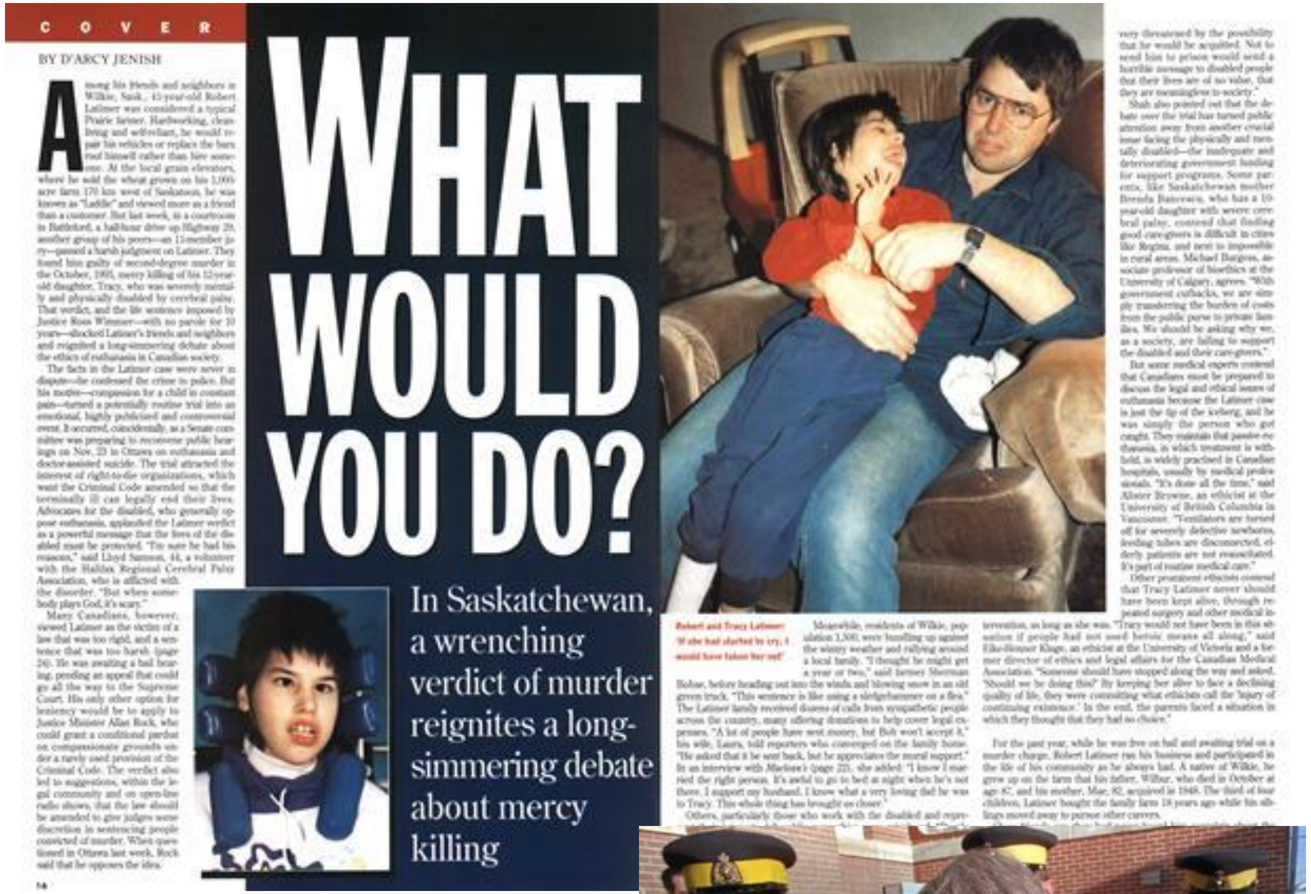
- Carbon monoxide displaces oxygen in the blood and deprives the heart, brain, and other vital organs of oxygen
- Carbon monoxide is colourless, odourless, and tasteless

Symptoms of carbon monoxide poisoning:

CO **Placement of CO(g) alarms**

- It is recommended that at least one alarm is installed on every level





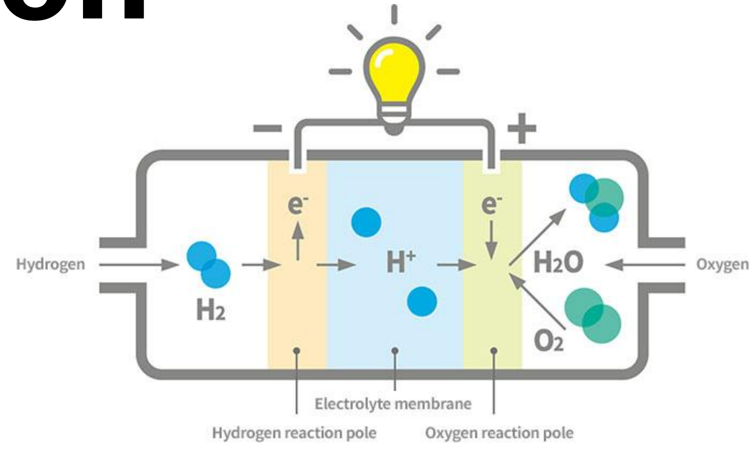
- October 24, 1993, Robert Latimer killed his 13-year old daughter Tracy Latimer by piping the exhaust fumes containing CO(g) into the interior of the car where she was placed
- Tracy had a severe form of cerebral palsy and suffered considerable pain
  - Father killed her to relieve her of her pain
- Triggered debates around health ethics and euthanasia

Other Forms of Combustion

Combustion of Hydrogen (Fuel Cells)

$2H_2(g) + O_2(g) \rightarrow 2H_2O(g) + energy$

dangerous



Combustion of Nitrogen-Containing Hydrocarbons

$C_xH_yN_z + O_2 \rightarrow CO_2(g) + H_2O(g) + N_2(g)$

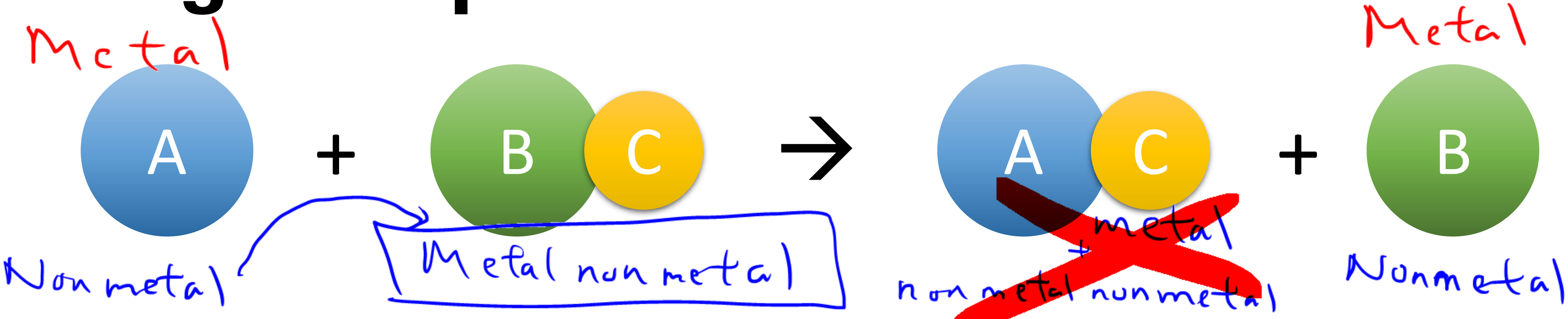
T

matchsticks

Combustion of Phosphorus

$P_4(s) + 5O_2(g) \rightarrow P_4O_{10}(g) + energy$

Single Displacement



- Metal displaces the metal (use the most common ionic charge for multivalent metals unless indicated otherwise)

$$\mathbf{Cu(s)} + 2\mathbf{Ag}NO_3(aq) \rightarrow \mathbf{Cu}(NO_3)_2(aq) + 2\mathbf{Ag(s)}$$

- Nonmetal displaces the nonmetal

$$\mathbf{F_2(g)} + 2K\mathbf{Cl}(aq) \rightarrow 2K\mathbf{F}(aq) + \mathbf{Cl_2}$$

Checkpoint

Predict the products of the following single displacement reactions and balance:

M M

a) 2 Fe + 3 Pb(NO_3)$_2$ → 2 Fe(NO_3)$_3$ + 3 Pb

b) Cl_2 + 2 NaI → 2 NaCl + I_2

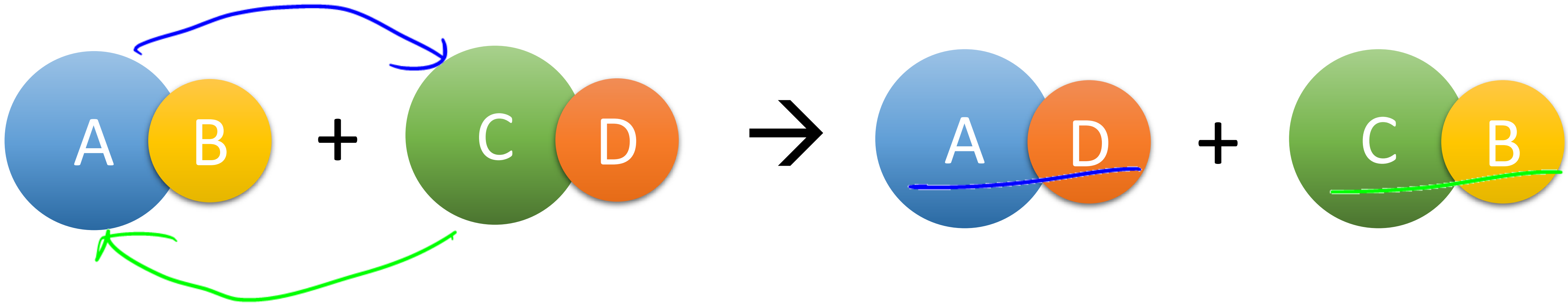
diatomic

H(OH)

c) 2 Li + 2 H_2O → 2 LiOH + H_2

2 4 2 2 2 2 2

Double Displacement



- Metals trade places or nonmetals trade places

  $AgNO_3(aq) + NaCl(aq) \rightarrow NaNO_3(aq) + AgCl(s)$

- Double displacement reactions occur if:
  - A solid precipitate forms OR H_2O forms OR gas forms

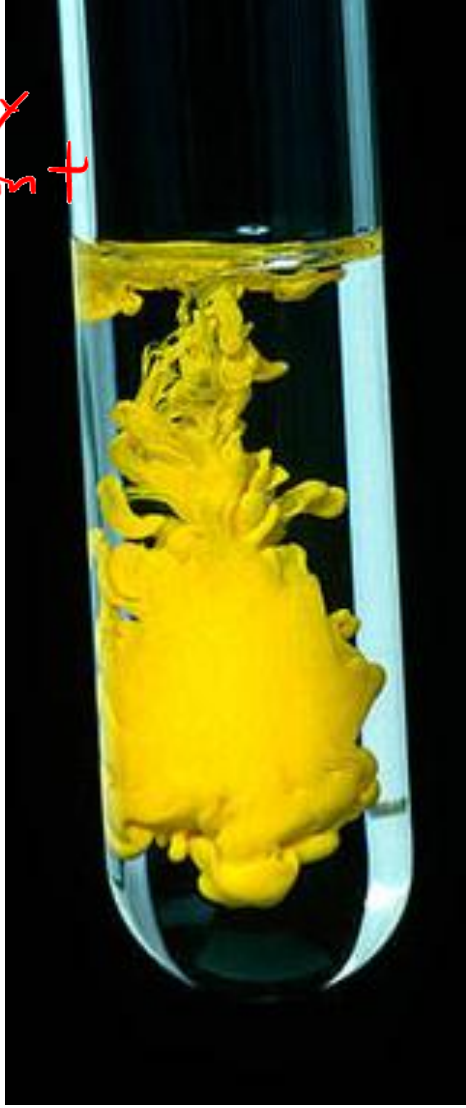
Gr II : K_{sp} solubility constant

Precipitate

- **Precipitate** is a solid formed from the reaction of two solutions; an insoluble compound

$$Pb(NO_3)_2(aq) + 2KI(aq) \rightarrow \mathbf{PbI_2(s)} + 2KNO_3(aq)$$

$$H_2O = H^+(OH)^-$$



$PbI_2(s)$ is a yellow precipitate

STOP Checkpoint STOP

Predict the products of the following double displacement reactions and balance:

A B C D A D C B

a) $CuS + 2KCl \rightarrow CuCl_2 + K_2S$

b) $CaBr_2 + 2KOH \rightarrow 2KBr + Ca(OH)_2$

belong to $CaBr_2$

c) $3Ca(OH)_2 + 2H_3(PO_4) \rightarrow Ca_3(PO_4)_2 + 6H_2O$

STOP Checkpoint STOP

$\cancel{Ca(OH)}$

$\cancel{Ca(OH)}$

$Ca(OH)_2$

Write the balanced chemical equation for the following and classify the type of reaction:

a) Silver nitrate + sodium phosphate → silver phosphate precipitate + sodium nitrate

$3 AgNO_3 + Na_3(PO_4) \longrightarrow Ag_3(PO_4) + 3 NaNO_3$

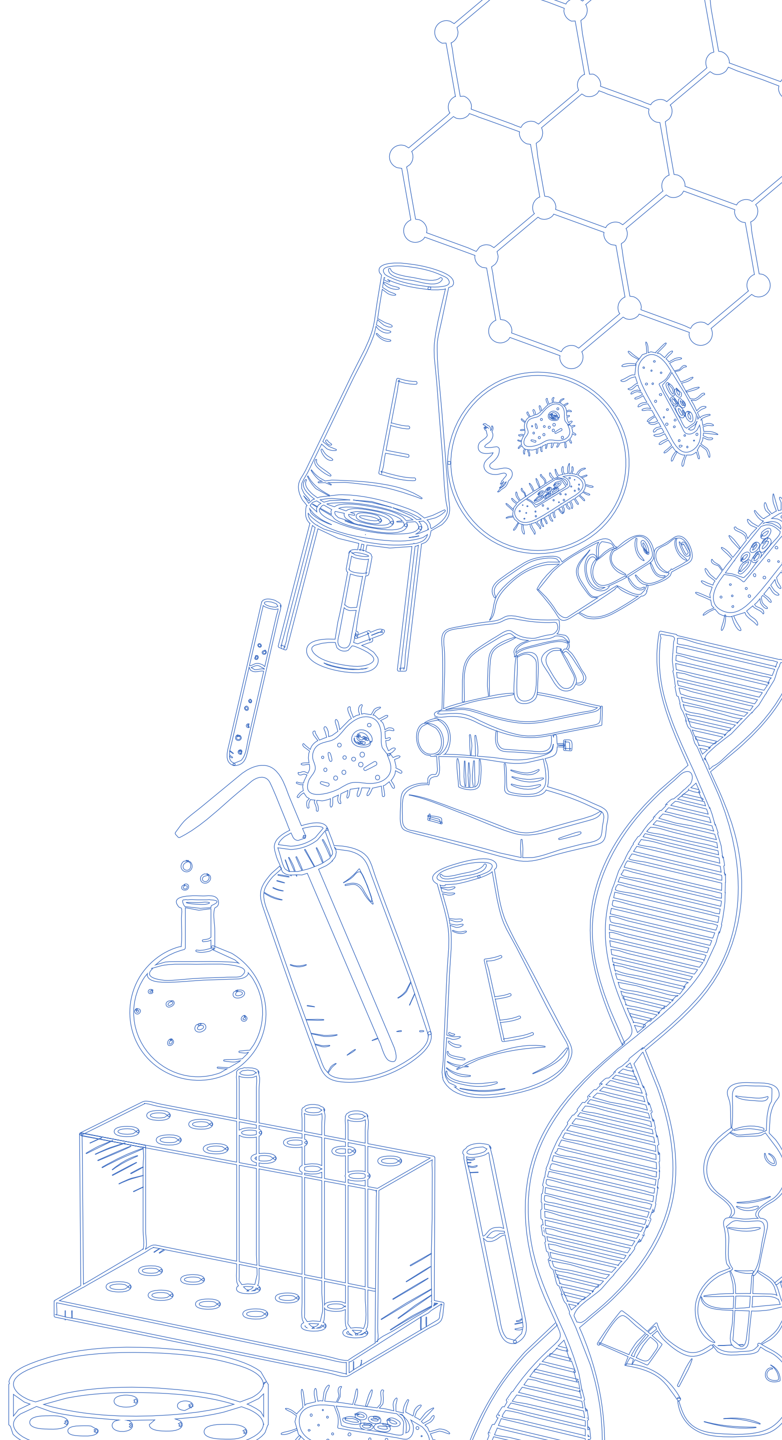
b) Calcium + water → hydrogen gas + calcium hydroxide

$1\ Ca + 2\ H(OH) \longrightarrow 1\ H_2 + 1\ Ca(OH)_2$

2 4 2 2 2

What I Learned Today:

- ❑ Law of Conservation of Mass
- ❑ Balancing Chemical Reactions
- ❑ Types of Chemical Reactions
  - Synthesis
  - Decomposition
  - Combustion (Complete and Incomplete)
  - Single Displacement
  - Double Displacement



Due next class: Class 2 Homework